National University of Computer and Emerging Sciences, Karachi

FAST School of Computing, Spring 2026

CS3001 – Computer Networks

1. Project Title and Group Information

Project Title:

IP Calculator for Subnet Design (Web-Based Network Planning Tool)

Group Members:

- Muneer Ahmed (21i-1179)
- Kapil Dev (22k-4942)

2. Proposed Project Description

Introduction

The proposed project is a **Web-Based IP Subnet Calculator for Network Design**, developed using **Python (Flask/Django) or Java**. The system will assist network administrators and organizations in efficiently designing and allocating subnets based on IP addressing requirements.

Subnetting is a fundamental concept in computer networks used to divide a large network into smaller logical sub-networks to improve performance, security, and address

management. Manual subnet calculations can be error-prone and time-consuming. This project aims to automate and simplify this process.

How the System Will Work

The system will operate as follows:

1. The user enters:
    a. An IPv4 address (e.g., 192.168.1.0)
    b. Subnet mask or CIDR notation (e.g., /24)
    c. (Optional) Required number of hosts for VLSM
2. The backend system will:
    a. Convert IP address into binary format.
    b. Apply subnetting rules using bitwise operations.
    c. Calculate network parameters.
    d. Validate input correctness.
3. The system will display:
    a. Network Address
    b. Broadcast Address
    c. First Usable Host
    d. Last Usable Host
    e. Total Number of Hosts
    f. Number of Usable Hosts
    g. Subnet Mask (Decimal and CIDR format)
4. For advanced mode (VLSM):
    a. The system will divide the network into optimized subnets based on required host counts.
    b. It will generate multiple subnet allocations automatically.

Functional Features

The project will include the following **functional features**:

1. IPv4 address validation and parsing
2. CIDR to subnet mask conversion

3. Subnet mask to CIDR conversion
4. Network address calculation
5. Broadcast address calculation
6. Host range calculation (first and last usable IP)
7. Total and usable host calculation
8. VLSM-based subnet allocation
9. Binary representation of IP and subnet mask
10. Error handling for invalid IP formats

Technologies to be Used

Option 1 (Recommended):

- Python
- Flask (Web Framework)
- HTML/CSS + Bootstrap

Option 2:

- Java
- JSP/Servlets or Spring Boot
- HTML/CSS

Expected Outcome

The final system will be a fully functional web application that allows organizations and students to:

- Design optimal subnet structures
- Reduce IP wastage
- Improve network efficiency
- Automate subnet planning

3. Plan of Work (Next 5 Weeks)

The team will divide responsibilities equally. Each member will contribute to specific **functional features** of the system.

Week 1: Requirement Analysis s Design

- Study subnetting and VLSM algorithms
- Design system architecture
- Prepare flowcharts and UI mockups

Week 2: Core Subnet Calculation Module

Member 1 (Muneer Ahmed):

- Implement IPv4 validation
- Implement CIDR conversion logic
- Develop network and broadcast address calculation

Member 2:

- Implement host range calculation
- Implement total/usable host calculation
- Implement subnet mask conversion

Week 3: Advanced Features (VLSM Implementation)

Member 1:

- Design VLSM allocation algorithm
- Implement dynamic subnet generation

Member 2:

- Implement binary representation module
- Add error handling and edge-case validation

Week 4: Integration s Testing

Member 1:

- Integrate backend subnet logic with frontend

Member 2:

- Perform functional testing for all subnet cases
- Validate calculations with manual examples

Week 5: Finalization s Documentation

- System testing
- Performance optimization
- Prepare final report
- Prepare presentation slides
- Demonstration preparation

References

1. Kurose, J. F., C Ross, K. W. (2017). *Computer Networking: A Top-Down Approach*. Pearson.
2. Tanenbaum, A. S., C Wetherall, D. J. (2011). *Computer Networks (5th Edition)*. Pearson.
3. RFC 950 – Internet Standard Subnetting Procedure.
4. RFC 4632 – Classless Inter-domain Routing (CIDR): The Internet Address Assignment and Aggregation Plan.
5. Python Software Foundation. *ipaddress — IPv4/IPvc manipulation library*.

c. Cisco Networking Academy. *Introduction to Subnetting and VLSM*